

SLURM Output Aggregator

Summary of `aggregate.py`

(a) Theory

`aggregate.py` contains no physics logic. It is a post-processing utility that collects JSON result files written by the SLURM job-array tasks of `mc_fast.py` and `mc_uniform_fast.py` and reproduces the same formatted tables that the standalone (sequential) versions of those scripts print. Its purpose is to allow individual L -value tasks — which run independently in parallel across cluster nodes — to be combined into a single coherent output after the job array completes.

(b) What the script does

The script operates in three modes, selected via `--mode`:

- **weighted**: reads `scaling_{L}.json` files produced by `mc_fast.py` SLURM tasks and prints the weighted-measure scaling study table ($\langle k \rangle / \sqrt{L}$, $\langle k \rangle / (\sqrt{L} \log L)$, $\sigma_k / L^{1/4}$, τ_{int} , acceptance rate, elapsed time).
- **uniform**: reads `mode_{L}.json` and `rld_{L}.json` files produced by `mc_uniform_fast.py` SLURM tasks and prints (a) the mode/mean convergence table for the uniform measure and (b) the per- L run-length distribution tables with BE and geometric comparisons and KL divergences. With `--paper`, also prints a side-by-side comparison of MC KL values against hardcoded paper reference values (see §(e) below).
- **both** (default): runs all three of the above, plus a cross-check table comparing weighted-measure and uniform-measure $\langle k \rangle$ at shared L values.

All output is printed to stdout. No CSV or other files are written.

(c) How the script does it

JSON result format

Each JSON file is a result dict written by a single SLURM task. The script reads each file and extracts the relevant fields:

- `scaling_{L}.json` (from `mc_fast.py`): keys `L`, `km`, `km_over_sql`, `ratio_logL`, `ksig_over_L14`, `tau`, `acc`, `elapsed`.
- `mode_{L}.json` (from `mc_uniform_fast.py`): keys `L`, `modal_k`, `modal_over_sql`, `ratio_logL`, `km`, `km_over_sql`, `ksig_over_L14`, `tau`, `elapsed`.
- `rld_{L}.json` (from `mc_uniform_fast.py`): keys `L`, `k_mean`, `r_bar`, `alpha`, `pred_k_gc`, `pred_r_gc`, `kl_be`, `kl_geom`, `kl_ratio`, `f_mc`, `f_be`, `f_geom` (the last three as string-keyed dicts of $j \rightarrow f(j)$ values).

L-value lists

The L values to aggregate are imported directly from the producing scripts: `SCALING_L_VALS` from `mc_fast.py` and `MODE_L_VALS`, `RLD_L_VALS` from `mc_uniform_fast.py`. This ensures the aggregator always uses exactly the same L list as the job array, with no hardcoded duplication. Missing result files are flagged with a warning rather than causing an error.

Directory defaults

Default result directories are `results_mc` (weighted) and `results_mc_uniform` (uniform), matching the `--outdir` defaults of the producing scripts. Both can be overridden via `--indir` (for single-mode use) or `--weighted-indir` / `--uniform-indir` (for `--mode both`).

Cross-check

`cross_check` identifies L values present in both `SCALING_L_VALS` and `MODE_L_VALS`, reads the corresponding JSON files from both result directories, and prints the ratio $\langle k \rangle_{\text{weighted}} / \langle k \rangle_{\text{uniform}}$. Since the weighted measure $\tilde{w}(\lambda) \propto k! 2^k / \prod_j m_j!$ strongly favours high- k partitions, this ratio should substantially exceed 1 at all L , confirming the distinction between the two ensembles discussed in §6.3 of the paper.

(d) Output produced

The script produces no files. All output is printed to stdout. The table formats mirror those of the producing scripts exactly, allowing the SLURM and sequential outputs to be compared line-by-line.

(e) One issue: stale `PAPER_KL` constant

The script contains a hardcoded constant:

```
PAPER_KL = {
    100: (0.040, 0.072),
    200: (0.023, 0.099),
    500: (0.026, 0.145),
}
```

These values are from a version of `mc_uniform_fast.py` that used the GC mean as the BE normaliser. The correct values in the current paper (§8.3, from `rld_fast.py` with canonical normaliser) are substantially different:

L	$D_{\text{KL}}(\cdot f_{\text{BE}})$ (current)	$D_{\text{KL}}(\cdot f_{\text{geom}})$ (current)
100	≈ 0.0018	≈ 0.080
200	≈ 0.0013	≈ 0.103
500	≈ 0.0010	≈ 0.147

The **--paper** flag should not be used until `PAPER_KL` is updated to reflect the current paper values. In the meantime, the plain `--mode uniform` output (without `--paper`) is correct and unaffected by this constant.